MINI AUTONOMOUS CAR WITH V2V DECISION MAKING

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

ELECTRICAL AND ELECTRONICS ENGINEERING

CHENNAI INSTITUTE OF TECHNOLOGY (AUTONOMOUS)
(Affiliated to Anna University, Chennai)
CHENNAI-600 069

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MARCH-2026

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BONAFIDE CERTIFICATE

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CO3	Ability to understand, formulate and propose new learning algorithms to solve engineering and societal problems of moderate complexity through multidisciplinary projects understanding commitment towards sustainable development.
CO4	Ability to demonstrate, prepare reports, communicate and work in a team as a member/leader by adhering to ethical responsibilities.
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Sustainable Development Goals (SDG)

SDG Number	SDG Title	Relevance with Project
SDG 3	Good Health and Well-being	The project directly contributes to improved patient care by enabling continuous real-time monitoring of vital parameters. This allows for timely detection of anomalies and faster medical intervention, thereby enhancing overall health outcomes.
SDG 9	Industry, Innovation, and Infrastructure	The project leverages innovative IoT technologies to integrate multiple sensors within a wearable glove. This approach fosters innovation in healthcare technology and strengthens the infrastructure for remote monitoring.
SDG 11	Sustainable Cities and Communities	By enabling remote patient monitoring, your system reduces the need for frequent hospital visits and supports smart healthcare practices. This contributes to building more resilient and sustainable communities, particularly in urban areas where healthcare resources can be optimized.
SDG 12	Responsible Consumption and Production	The project emphasizes an energy-efficient design through the use of low-power sensors and optimized power management strategies. This not only reduces energy consumption but also supports sustainable production practices in the development of medical devices.

ACKNOWLEDGEMENT

We express our gratitude to our Chairman **Shri.P. SRIRAM** and all trust members of Chennai institute of technology for providing the facility and opportunity to do this project as a part of our undergraduate course.

We are grateful to our Principal **Dr.A. RAMESH M.E**, **Ph.D.** for providing us the facility and encouragement during the course of our work.

We sincerely thank our Head of the Department, **Dr.R.SANKAR M.E,Ph.D.,** Department of Electrical and Electronics Engineering for having provided us valuable guidance, resources and timely suggestions throughout our work.

We sincerely thank our Project Guide, **Mr.M. EZHILMARAN M.E.,** Professor, Department of Electrical and Electronics Engineering for having provided us with valuable guidance, resources and timely suggestions throughout our work.

We would like to extend our thanks to our **Project coordinator of the Department of Electrical and Electronics Engineering**, for their valuable suggestions throughout this project.

We wish to extend our sincere thanks to all **Faculty members of the Department of Electrical and Electronics Engineering** for their valuable suggestions and their kind cooperation for the successful completion of our project.

We wish to acknowledge the help received from the **Lab Instructors of the Department of Electrical and Electronics Engineering** and others for providing valuable suggestions and for the successful completion of the project.

ABSTRACT

Advancements in embedded systems, wireless communication, and autonomous technologies are transforming modern transportation systems. One of the major challenges in road safety occurs at intersections, where vehicles approach from different directions with limited coordination, often leading to collisions due to delayed human reactions, poor visibility, or lack of traffic control infrastructure. Traditional traffic management systems such as traffic signals and manual control rely heavily on centralized infrastructure and human compliance, which may not always ensure safe and efficient traffic flow.

This project presents the design and development of a low-cost miniature autonomous vehicle system with Vehicle-to-Vehicle (V2V) communication capable of intelligent decision-making at road intersections. The system consists of two miniature autonomous vehicles developed using embedded system technologies. Each vehicle is equipped with an ESp-32 microcontroller, ultrasonic sensors, DC motors with motor drivers, and wireless communication modules. The sensors detect nearby vehicles and intersection ones, while the wireless module enables real- time communication between vehicles to exchange information such as vehicle ID, distance, and movement status.

To further validate the system, a digital simulation environment using VEINS (OMNeT++ integrated with SUMO) was used to model traffic behavior and wireless communication between vehicles. The experimental results demonstrate that the proposed system successfully enables autonomous negotiation and collision-free intersection crossing through cooperative communication and real-time decision making. This project highlights how intelligent transportation concepts can be implemented using affordable embedded hardware and simple algorithms, providing a practical platform for research in autonomous systems and smart mobility technologies.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

A. SYMBOLS

Symbol	Description
(V_i)	Vehicle (i) in the autonomous vehicle system
(V_j)	Neighboring vehicle involved in communication
(D_i)	Distance of vehicle (i) from the intersection
(D_s)	Distance measured by ultrasonic sensor
(S_i)	Speed of vehicle (i)
(ID)	Unique identification number of a vehicle
(T_d)	Decision time taken by the system
(P_i)	Priority assigned to vehicle (i)
(Z)	Intersection decision zone
(R_c)	Communication range between vehicles
(C_{msg})	Communication message exchanged between vehicles
(T_{cycle})	System decision cycle time
(GO)	Command indicating the vehicle can proceed
$(WAIT)$	Command indicating the vehicle must stop
(V_{cc})	Supply voltage of the embedded system
(I_{total})	Total current consumption of the system
(P_{total})	Total power consumption of the system
(f_{clk})	Microcontroller clock frequency
(t_{delay})	System response delay
(d_{safe})	Safe distance maintained between vehicles
(N_v)	Number of vehicles in the system

B. ABBREVIATIONS

Abbreviation	Expanded Form
V2V	Vehicle-to-Vehicle Communication
ITS	Intelligent Transportation System
MCU	Microcontroller Unit
IR	Infrared Sensor
DC	Direct Current
PWM	Pulse Width Modulation
UART	Universal Asynchronous Receiver Transmitter
I2C	Inter-Integrated Circuit Communication
SPI	Serial Peripheral Interface
GPI	General Purpose Input Output
IOT	Internet Of Things
BLE	Bluetooth Low Energy

C. NOMENCLATURE

Term	Definition
Autonomous Vehicle	Vehicle that moves without human control using sensors and a controller
ESP32 Controller	Microcontroller used for vehicle control and Bluetooth communication
Vehicle-to-Vehicle Communication	Data exchange between vehicles using Bluetooth
Decision Algorithm	Logic used to decide vehicle movement (go or slow down)

Safe Distance	Minimum distance maintained between vehicles

Ultrasonic Sensor	Sensor used to measure distance to nearby vehicles or obstacles
Bluetooth Module	Wireless interface in ESP32 used for communication between cars
Intersection Zone	Area where vehicles from different directions meet
Vehicle Priority	Rule used to decide which vehicle moves first
Control Signal	Command sent from ESP32 to control vehicle movement
Collision Avoidance	Method used to prevent vehicles from crashing

CHAPTER:1

INTRODUCTION

The rapid growth of intelligent transportation technologies has created opportunities to improve road safety and traffic efficiency through automation and communication between vehicles. Traditional transportation systems rely heavily on human drivers to make decisions while driving. However, human errors such as delayed reaction time, lack of situational awareness, and poor judgment are major causes of road accidents. As a result, researchers and engineers are increasingly exploring autonomous systems and cooperative vehicle technologies that can assist or replace human decision-making in certain driving situations. Autonomous vehicles represent a major advancement in modern transportation systems. These vehicles are capable of sensing their environment, communicating with other vehicles, and making real-time decisions based on available data. One of the key concepts enabling this technology is Vehicle-to-Vehicle (V2V) communication, where vehicles exchange information such as speed, position, and direction. By sharing this information, vehicles can coordinate their movement and avoid potential collisions[3,4].

This project focuses on developing a mini autonomous car system with V2V decision making using embedded system technology. The system uses an ESP32 microcontroller, which includes a built-in Bluetooth communication module. The ESP32 enables vehicles to exchange information wirelessly and make cooperative decisions. Each vehicle is equipped with sensors to detect nearby vehicles or obstacles, while a motor driver module controls the movement of the vehicle.[5] In each case, the vehicles communicate and apply a simple decision algorithm to avoid collisions and maintain safe movement.The project highlights how low-cost embedded systems and wireless communication can be used to implement fundamental concepts of intelligent transportation systems. By combining sensing, communication, and decision-making, the system provides a practical demonstration

1.1 Background of the Study

Transportation systems are evolving with the integration of embedded systems, wireless communication technologies, and intelligent control algorithms. Autonomous vehicle technology aims to reduce accidents, improve traffic flow, and enhance overall transportation efficiency. One of the important components of autonomous vehicle systems is the ability of vehicles to communicate with each other in real time.

Vehicle-to-Vehicle communication allows vehicles to exchange information such as location, speed, and driving intentions. This communication enables vehicles to make cooperative decisions, especially in situations where collision risks are high. In recent years, microcontrollers with integrated wireless capabilities, such as the ESP32, have made it possible to develop small-scale prototypes that demonstrate these concepts.

1.2 Problem Statement

Road accidents frequently occur due to lack of coordination between vehicles, especially in situations where vehicles approach each other from different directions or when one vehicle follows another too closely. Human drivers may fail to react quickly enough to avoid collisions. This lack of communication can lead to dangerous situations such as head-on collisions, rear-end collisions, or intersection crashes.

1.3 Motivation

The motivation behind this project is to explore how simple embedded systems and wireless communication can be used to implement cooperative vehicle behavior. By enabling vehicles to exchange information and make basic decisions automatically, it is possible to improve safety and reduce the chances of accidents. Using an ESP32

microcontroller with built-in Bluetooth communication provides a cost-effective platform for demonstrating Vehicle-to-Vehicle communication. The project also aims to create a practical model that illustrates how autonomous vehicles can detect nearby vehicles, communicate their presence, and adjust their movement accordingly.

1.4 Objective of the Work

The main objectives of this project are:

1. To design and develop a mini autonomous vehicle system using the ESP32 microcontroller.
2. To implement Bluetooth-based Vehicle-to-Vehicle communication between vehicles.
3. To develop a decision algorithm that enables vehicles to coordinate their movement.
4. To control vehicle motion using a motor driver module.
5. To demonstrate collision avoidance in three test scenarios: opposite direction movement, same direction movement, and perpendicular intersection movement.

1.5 Scope of the Study

This project focuses on developing a prototype autonomous vehicle system capable of basic communication and decision-making. The system demonstrates how vehicles can exchange information using Bluetooth and coordinate their movement using embedded algorithms.[7]

The project is limited to small-scale autonomous vehicle models controlled by ESP32 microcontrollers. The results provide a foundation for further research and development in intelligent transportation systems and cooperative autonomous vehicle technologies.

CHAPTER:2
LITERATURE REVIEW

The development of autonomous vehicles and intelligent transportation systems has attracted significant attention from researchers and engineers in recent years. Advances in embedded systems, wireless communication, and sensor technologies have enabled vehicles to communicate with each other and make cooperative decisions to improve road safety. Vehicle-to-Vehicle (V2V) communication plays a crucial role in enabling autonomous vehicles to exchange information such as speed, position, and driving intentions. This chapter reviews relevant research works related to autonomous vehicles, V2V communication technologies, embedded system platforms, and collision avoidance systems.[2,8]

2.1 Autonomous Vehicles and Intelligent Transportation Systems

Autonomous vehicles are designed to operate with minimal or no human intervention by sensing the surrounding environment and making real-time decisions. These systems combine sensors, communication modules, and embedded controllers to ensure safe navigation. Intelligent Transportation Systems (ITS) integrate advanced technologies into transportation infrastructure and vehicles to improve traffic efficiency and safety.[30]

Research in autonomous vehicles focuses on enabling vehicles to detect obstacles, communicate with other vehicles, and make intelligent decisions to avoid collisions. Various technologies such as radar, cameras, LiDAR, and ultrasonic sensors are commonly used to detect nearby vehicles and obstacles. Therefore, low-cost embedded platforms combined with simple sensors have become popular for educational and experimental purposes.[29,2]

2.2 Vehicle-to-Vehicle Communication Technologies

Vehicle-to-Vehicle communication enables vehicles to exchange informationdirectly without relying on centralized infrastructure. Through V2V communication, vehicles can share data such as position, speed, and direction, allowing them to coordinate movement and reduce the risk of accidents.

Several wireless technologies have been used for V2V communication, including Dedicated Short Range Communication (DSRC), Wi-Fi, Zigbee, and Bluetooth.The ESP32 includes integrated Bluetooth and Wi-Fi modules, making it an ideal platform for implementing V2V communication in embedded systems. By using Bluetooth communication, vehicles can transmit and receive information about their movement and adjust their behavior accordingly.

2.3 Embedded Systems for Autonomous Vehicles

Embedded systems play a critical role in autonomous vehicle operation by processing sensor data and controlling vehicle movement. A microcontroller acts as the central processing unit that collects data from sensors, runs decision algorithms, and sends control signals to actuators such as motors.

Modern embedded platforms such as the ESP32 provide powerful processing capabilities along with integrated communication features.Motor driver modules are commonly used in embedded systems to control the direction and speed of DC motors. By using Pulse Width Modulation (PWM) signals generated by the microcontroller, the motor driver can regulate vehicle speed and movement.[9] The combination of sensors, microcontroller processing, and motor control enables the vehicle to respond dynamically to changes in its environment.[11,18]

2.4 Related Implementations

Several research projects and experimental prototypes have been developed to demonstrate autonomous vehicle communication and collision avoidance systems. Many of these systems focus on scenarios such as vehicles approaching from opposite directions, vehicles following each other in the same lane, and vehicles arriving at intersections simultaneously.

In some implementations, sensors are used to detect obstacles while wireless communication modules allow vehicles to exchange information about their movement. Based on this information, vehicles can apply simple decision algorithms to determine whether to move, slow down, or stop.[11,23]

Small-scale autonomous vehicle models are commonly used in academic research to study cooperative vehicle behavior. These models provide a practical platform for testing communication protocols and decision-making algorithms in controlled environments. By analyzing different traffic scenarios, researchers can evaluate the effectiveness of collision avoidance strategies and improve the reliability of autonomous vehicle systems.[8,17]

CHAPTER 3
METHODOLOGY

This chapter explains the design and implementation approach used to develop the mini autonomous vehicle system with Vehicle-to-Vehicle communication. The methodology focuses on integrating embedded hardware, wireless communication, and a decision-making algorithm to enable vehicles to coordinate their movement and avoid collisions.[25,3] The system uses an ESP32 microcontroller with built-in Bluetooth communication to allow vehicles to exchange information and make cooperative decisions in real time.

The development process includes hardware design, communication setup, algorithm implementation, and testing under different vehicle interaction scenarios.

3.1 System Overview

The proposed system consists of miniature autonomous vehicles controlled by an ESP32 microcontroller. Each vehicle is equipped with sensors to detect nearby vehicles or obstacles and a motor driver module to control the movement of the vehicle. The vehicles continuously exchange information regarding their movement and position. 19,33]

The overall system consists of four main components:

1. Sensing Unit – Detects nearby vehicles or obstacles.
2. Communication Unit – Exchanges information between vehicles using Bluetooth.
3. Decision-Making Unit – Processes data and determines vehicle behavior.
4. Control Unit – Controls motor movement through the motor driver module.

Table 3.1: System Hardware Components

S. No	Component	Quantity	Description
1	ESP32 Microcontroller	1	Main controller responsible for processingsensor data and Bluetooth communication
2	Ultrasonic Sensor (HC- SR04)	1	Measures distance between vehicles and detects nearby obstacles
3	Motor Driver Module (L298N)	1	Controls direction and speed of DC motors
4	DC Gear Motors	2	Provides movement to the autonomous vehicle
5	Servo Motor (SG90)	1	Rotates the ultrasonic sensor for scanning nearby areas
6	Chassis with Wheels	1	Mechanical structure of the vehicle
7	Battery Pack	1	Power supply for ESP32, sensors, and motors
8	Bluetooth Module (Built-in ESP32)	1	Enables wireless communication between vehicles

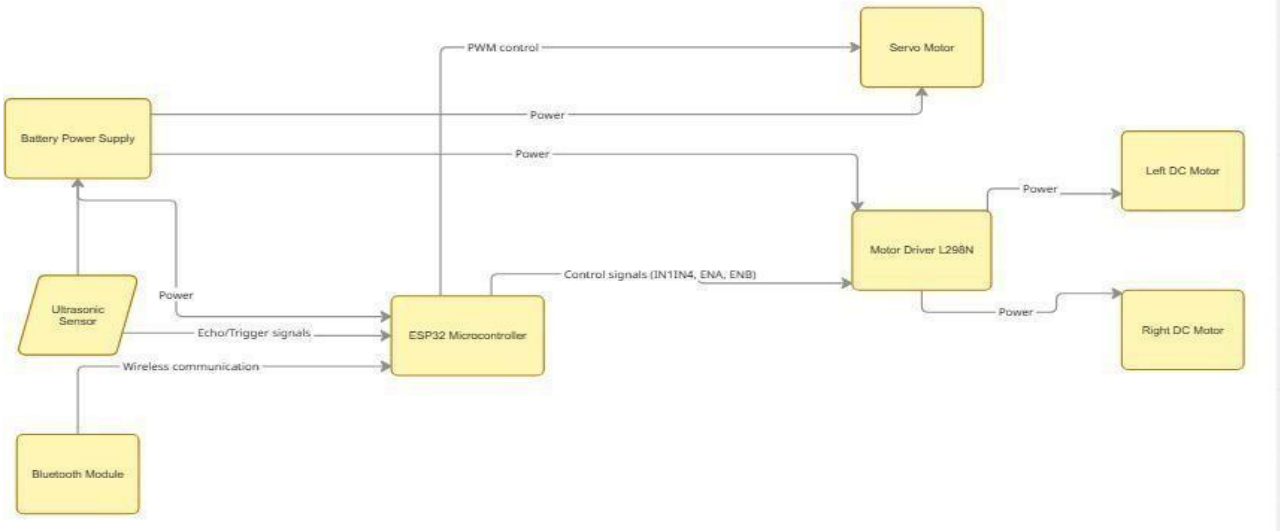

Figure 3.1 : Mini Autonomous Vehicle Prototype

3.2 Hardware Design Development

The hardware architecture of the system is designed using compact embedded components that enable sensing, communication, and vehicle movement control. The main hardware components used in the system include:

ESP32 Microcontroller – Acts as the central controller of the vehicle. It processes sensor inputs, manages Bluetooth communication, and executes the decision algorithm.

1) Ultrasonic Sensor – Used to measure the distance between vehicles and detect nearby obstacles.
2) Motor Driver Module – Controls the direction and speed of the DC motors that drive the vehicle.
3) DC Motors – Provide motion for the autonomous vehicle.
4) Power Supply – Provides the required voltage to operate the microcontroller and motors.

The ESP32 microcontroller receives sensor data, communicates with other vehicles through Bluetooth, and sends control signals to the motor driver module to regulate vehicle movement.

Table 3.2: Sensor Specifications

Parameter	Ultrasonic Sensor (HC-SR04)
Operating Voltage	5V
Operating Current	15 mA
Measuring Range	2 cm – 400 cm
Measuring Angle	15°
Accuracy	±3 mm
Trigger Input Signal	10 µs TTL Pulse
Echo Output Signal	Pulse Width proportional to distance

3.3 Sensor Integration

Sensors play an important role in detecting the surrounding environment of the vehicle. In this project, ultrasonic sensors are used to measure the distance between vehicles and detect nearby obstacles.[31,28] The ultrasonic sensor works by transmitting sound waves and measuring the time taken for the echo to return. Based on this time delay, the distance between the vehicle and an object can be calculated.

The sensor continuously monitors the environment and sends distance information to the ESP32 microcontroller. If the detected distance falls below a predefined safe threshold, the controller adjusts the vehicle's movement to maintain safe spacing between vehicles.

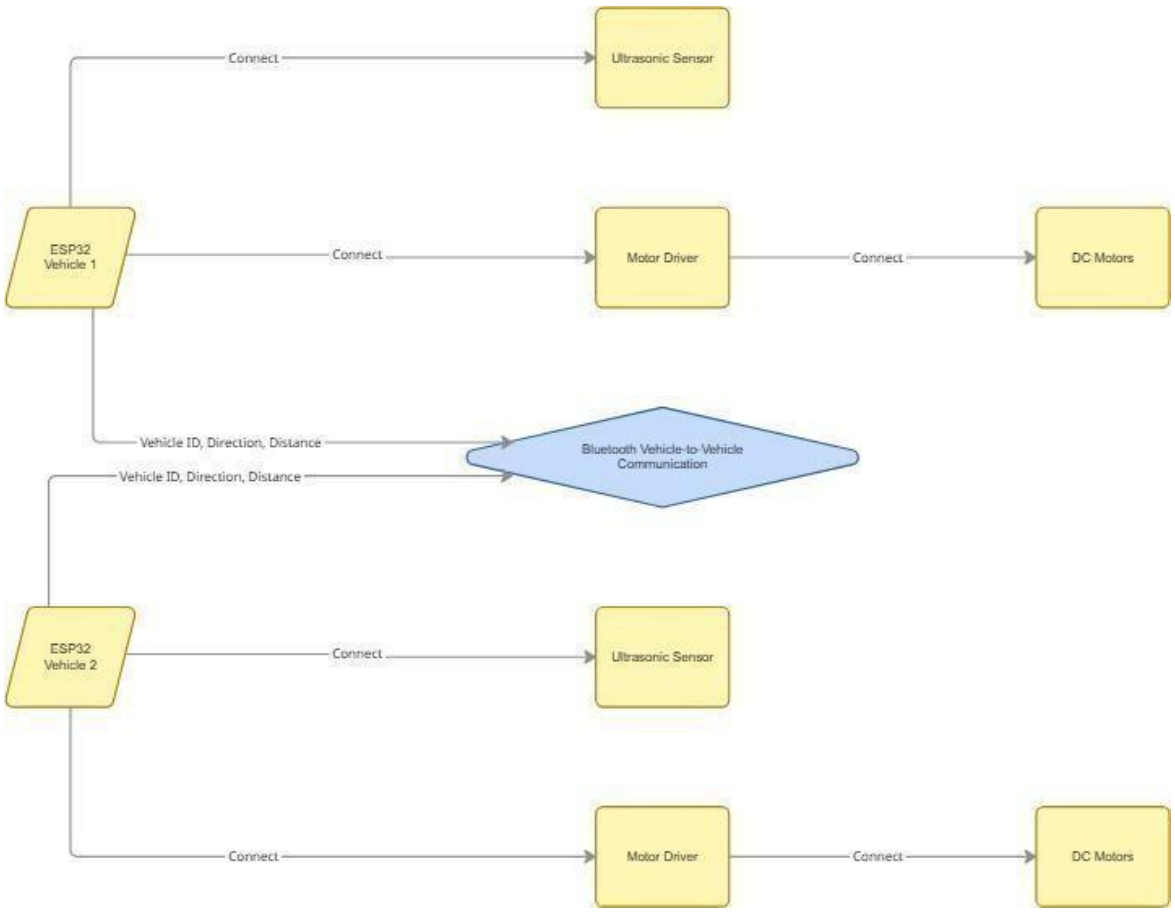

Figure 3.2: Wireless Communication Setup

3.4 Wireless Communication Implementation

Wireless communication between vehicles is implemented using the Bluetooth capability of the ESP32 microcontroller. The ESP32 has an integrated Bluetooth module that allows vehicles to transmit and receive information without additional hardware. Each vehicle broadcasts information such as:

1) Vehicle identification number
2) Distance from nearby vehicles
3) Movement status (moving or stopped)

This information is received by neighboring vehicles through Bluetooth communication. The communication process occurs continuously, ensuring that vehicles always have updated information about nearby vehicles. Using Bluetooth communication simplifies the system design and enables reliable data exchange between vehicles in short-range environments.[14,22]

3.5 Implementation Flow

The implementation of the system follows a step-by-step process to ensure reliable operation.

1. The ESP32 microcontroller initializes all sensors and communication modules.
2. The ultrasonic sensor measures the distance to nearby vehicles or obstacles.
3. The ESP32 transmits its current status to other vehicles using Bluetooth.
4. Each vehicle receives information from neighboring vehicles.
5. The decision algorithm analyzes sensor data and received communication data.
6. Based on the analysis, the vehicle determines whether to move, slow down, or stop.
7. Control signals are sent to the motor driver module to adjust vehicle movement.
8. This process repeats continuously during vehicle operation.

3.6 Decision-Making Algorithm

1) The decision-making algorithm is responsible for determining how each vehicle should behave based on sensor data and communication information. The algorithm uses simple rule-based logic to ensure safe vehicle movement.
2) If another vehicle is detected within the safe distance range, the vehicle slows down or stops.
3) If no obstacle or vehicle is detected within the safety range, the vehicle continues moving forward.
4) When multiple vehicles approach the same area, priority rules determine which vehicle moves first.
5) This rule-based approach ensures predictable system behavior and reduces the risk of collisions.

3.7 Performance Evaluation

The system was evaluated through practical testing using three main scenarios that simulate common traffic situations.

A).Scenario 1 – Opposite Direction Movement

Two vehicles approach each other from opposite directions. The system ensures that the vehicles maintain a safe distance and avoid collision.

B).Scenario 2 – Same Direction Movement

One vehicle follows another vehicle moving in the same direction. The following vehicle adjusts its speed based on the distance to the leading vehicle.

C).Scenario 3 – Intersection Movement

Two vehicles approach an intersection from perpendicular directions. The system determines vehicle priority and allows one vehicle to proceed while the other waits.

These test cases demonstrate the effectiveness of the system in enabling cooperative vehicle behavior and collision avoidance.

CHAPTER 4
ARCHITECTURE AND DESIGN

This chapter explains the structural design and operational architecture of the proposed autonomous vehicle system. The architecture integrates sensing, communication, decision-making, and motion control components into a single embedded system. The design focuses on enabling vehicles to detect nearby vehicles, exchange information through wireless communication, and make cooperative decisions to avoid collisions.

The system architecture is built around the ESP32 microcontroller, which acts as the central control unit. The ESP32 processes sensor inputs, communicates with nearby vehicles through Bluetooth, and generates control signals to regulate vehicle movement through the motor driver module.

4.1 Overall Architecture

The overall architecture of the system consists of four main modules:

1. Sensing Module
2. Communication Module
3. Decision-Making Module
4. Control and Actuation Module

The sensing module detects nearby vehicles or obstacles using ultrasonic sensors. These sensors continuously monitor the distance between vehicles and provide real- time information to the microcontroller. The communication module uses the built-in Bluetooth capability of the ESP32 to enable Vehicle-to-Vehicle communication. Vehicles exchange information such as vehicle identification, distance measurements, and movement status.The decision-making module processes both sensor data and received communication data. Based on predefined decision rules,

the system determines the appropriate action for the vehicle, such as continuing movement, slowing down, or stopping. The control and actuation module executes the decision by controlling the motor driver module. This modular architecture allows the system to operate reliably while maintaining flexibility for future improvements.

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4.2 Sensor And Communication Architecture

The sensor and communication architecture forms the perception and interaction layer of the system. Sensors are responsible for detecting the physical environment, while the communication module allows vehicles to share information with each other. The ultrasonic sensor measures the distance between the vehicle and nearby objects. The sensor transmits ultrasonic waves and measures the time taken for the echo to return. The ESP32 calculates the distance based on this time delay and determines whether another vehicle is within the safe range.

The Bluetooth communication system enables vehicles to exchange information wirelessly. Each ESP32 periodically broadcasts data that includes the vehicle ID and movement status. Other vehicles receive this information and update their internal system state accordingly. By combining sensor data and communication data, the system achieves a more reliable understanding of the surrounding environment.[19,20]

4.3 Vehicle Control System

The vehicle control system manages the physical movement of the autonomous vehicle. The control system includes the ESP32 microcontroller, motor driver module, and DC motors.

The motor driver module acts as an interface between the microcontroller and the motors. Since the ESP32 cannot supply enough current to drive motors directly, the motor driver amplifies control signals and provides the required power to the motors.

The ESP32 generates Pulse Width Modulation (PWM) signals that control the speed of the motors. By adjusting the duty cycle of the PWM signal, the system can regulate the speed of the vehicle.

Based on the decision-making algorithm, the ESP32 sends control commands to the motor driver to perform actions such as:

1. Move forward
2. Slow down
3. Stop the vehicle

This control mechanism ensures smooth and stable vehicle movement.

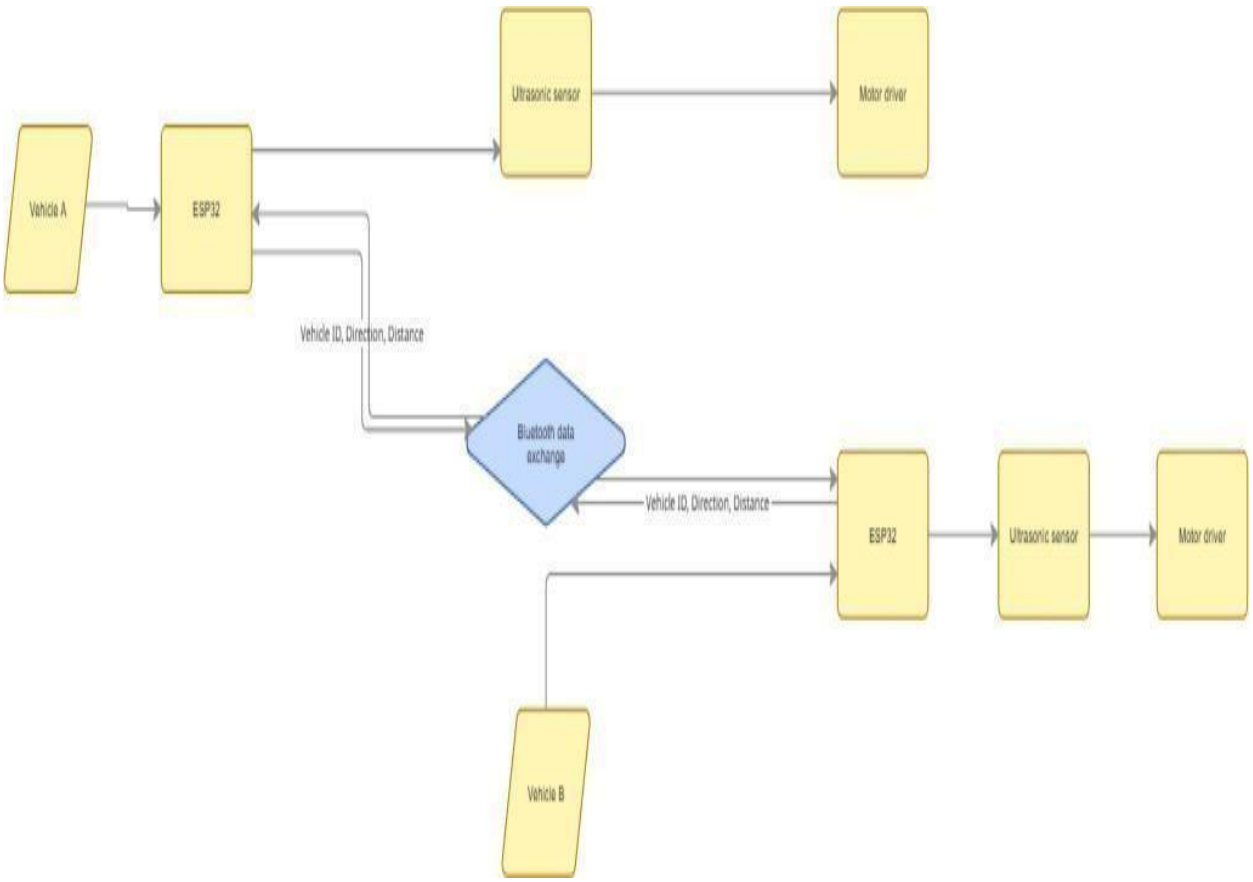

Figure 4.3: Vehicle-to-Vehicle Communication Model

4.4 Implementation Notes (Embedded System Flow)

The implementation of the system follows a structured embedded system workflow. First, the ESP32 initializes all hardware components, including sensors and communication modules. Once the system starts operating, the ultrasonic sensor continuously measures the distance to nearby vehicles or obstacles. At the same time, the ESP32 sends and receives Bluetooth messages from other vehicles. These messages provide information about the presence and movement of neighboring vehicles. The microcontroller processes both sensor data and communication data using the decision algorithm. Based on the results of this analysis, the system determines the appropriate action for the vehicle.[15,35]

Finally, control signals are sent to the motor driver module to execute the decision. This process runs continuously in a loop, allowing the vehicle to react dynamically to changes in its environment. This architecture demonstrates how sensing, communication, and control can be integrated into a cooperative autonomous vehicle system.

CHAPTER 5
SOFTWARE AND ALGORITHM IMPLEMENTATION

This chapter describes the software design and algorithm used to control the autonomous vehicle system. The software is responsible for managing communication between vehicles, processing sensor data, and controlling vehicle movement. The program is implemented using the ESP32 microcontroller, which integrates sensing, communication, and control functions within a single embedded platform. The program controls the operation of sensors, manages Bluetooth communication between vehicles, and executes the decision-making algorithm that determines vehicle behavior.[17,29]

5.1 Microcontroller Programming

The ESP32 microcontroller acts as the central processing unit of the system. It receives input from sensors, processes communication data from other vehicles, and generates control signals for the motor driver module.

The main control loop of the program performs the following tasks:

1. Read sensor data from the ultrasonic sensor
2. Send vehicle status information using Bluetooth communication.
3. Receive communication data from other vehicles.
4. Execute the decision-making algorithm.
5. Control the vehicle movement through the motor driver.

This continuous cycle allows the system to respond dynamically to changes in the surrounding environment.

Table 5.1: Vehicle Communication Data Structure

Field	Description	Example Value
Vehicle ID	Unique identifier assigned to each vehicle	1
Direction	Current movement direction of vehicle	F (Forward)
Distance	Distance measured by ultrasonic sensor	18 cm
Status	Current vehicle action	Moving / Stop

5.2 V2V Communication Protocol

Vehicle-to-Vehicle communication is implemented using the Bluetooth capability of the ESP32 microcontroller. Each vehicle broadcasts its presence and movement status to nearby vehicles.

The communication data transmitted between vehicles includes:

1) Vehicle identification number
2) Distance information from sensors
3) Current movement status

Bluetooth communication allows vehicles to exchange information in real time. This data helps vehicles understand the presence and behavior of other vehicles in the system. The communication protocol ensures that vehicles continuously update their status and receive updates from neighboring vehicles, enabling cooperative decision-making.[1,2]

5.3 Collision Avoidance Algorithm

The collision avoidance algorithm is designed to prevent vehicles from colliding by controlling their movement based on sensor readings and communication data. The algorithm follows simple rule-based logic to determine vehicle behavior. When the system detects another vehicle within a predefined safe distance, the vehicle slows down or stops. If the path is clear, the vehicle continues moving forward.

The algorithm evaluates the environment continuously and ensures that vehicles maintain a safe distance from each other. By combining sensor detection and communication information, the system improves decision accuracy. The decision logic ensures that vehicles do not move into potentially dangerous situations, thereby reducing the risk of collisions.[4,13]

5.4 System Integration

The complete system integrates hardware components with the software algorithm to create a functional autonomous vehicle platform. The ESP32 microcontroller serves as the central hub connecting sensors, communication modules, and motor drivers. Sensor data provides environmental awareness, while Bluetooth communication enables interaction between vehicles. This integration of sensing, communication, and control allows the system to operate as a cooperative autonomous vehicle network.

5.5 Experimental Setup

The experimental setup consists of miniature vehicles equipped with ESP32 controllers, ultrasonic sensors, and motor driver modules. Each vehicle operates independently but communicates with other vehicles using Bluetooth. The vehicles are placed in controlled environments to simulate different traffic scenarios. During the experiments, vehicles exchange information and apply the decision algorithm to coordinate their movement.[16,19]

5.6 Integration with Hardware

The ESP32 microcontroller integrates sensors, Bluetooth communication, and the motor driver module. Sensor data and Bluetooth messages are processed by the ESP32 to determine vehicle movement. Based on the decision algorithm, control signals are sent to the motor driver to move, slow down, or stop the vehicle. This integration enables

This integration enables coordinated autonomous operation.

5.7 Observation

The system was tested under different scenarios to observe vehicle behaviour and evaluate system performance. The vehicles successfully communicated with each other and adjusted their movement based on the received information. The observations show that the vehicles maintained safe distances and avoided collisions in all tested scenarios. The Bluetooth communication provided reliable data exchange between vehicles, and the decision algorithm effectively controlled vehicle movement.

The results confirm that the proposed system can demonstrate basic autonomous vehicle coordination using embedded systems and wireless communication.

CHAPTER 6
VERIFICATION AND TESTING

Verification and testing are important steps in ensuring that the developed autonomous vehicle system functions correctly and safely. This chapter describes the testing procedures used to evaluate the performance of the system. The tests were conducted to verify hardware functionality, communication reliability, and the effectiveness of the decision-making algorithm.

The system was tested under controlled conditions using miniature vehicles equipped with ESP32 controllers, sensors, and motor driver modules. The testing process focused on verifying that the vehicles could communicate with each other, detect nearby vehicles, and make correct movement decisions to avoid collisions.

6.1 Hardware Testing methodology

The first stage of testing involved verifying the correct operation of the hardware components used in the system. The ESP32 microcontroller was tested to ensure proper power supply and program execution. The ultrasonic sensors were tested to confirm that they could accurately measure distance between vehicles. [35]

6.2 Communication Testing

Communication testing was performed to verify the reliability of Bluetooth communication between vehicles. Each vehicle was programmed with a unique identification and configured to transmit and receive data through the ESP32 Bluetooth module. The tests confirmed that Bluetooth communication provided stable and reliable data exchange within the operating range of the vehicles.[4,5]

6.3 Vehicle Coordination Testing

Vehicle coordination testing was conducted to verify that the decision-making algorithm correctly controlled vehicle movement. The vehicles were placed in different positions and allowed to communicate with each other through Bluetooth. The system monitored the sensor readings and communication messages to determine the appropriate action for each vehicle. [9,27]

Based on the algorithm, the vehicles adjusted their movement to maintain safe distance and avoid collisions. The tests confirmed that the vehicles were able to coordinate their actions effectively and respond correctly to different situations.

6.4 Test Case Validation

The system was validated using three main test scenarios that represent common vehicle interaction situations.

Test-Case 1 Opposite Direction Movement

Two vehicles approach each other from opposite directions. The system detects the approaching vehicle and ensures that both vehicles maintain a safe distance to prevent collision.

Test-Case 2 Same Direction Movement

One vehicle follows another vehicle moving in the same direction. The following vehicle detects the vehicle ahead and reduces speed to maintain safe spacing.

Test-Case 3 Intersection Scenario

Two vehicles approach an intersection from perpendicular directions. Using communication and sensor data, the system determines which vehicle should proceed first while the other vehicle waits. The results of these tests demonstrate that the proposed system successfully performs cooperative decision-making and collision avoidance.[3]

CHAPTER 7

RESULTS

This chapter presents the results obtained from testing the autonomous vehicle system developed in this project. The experiments were conducted using miniature vehicles equipped with ESP32 microcontrollers, ultrasonic sensors, Bluetooth communication, and motor driver modules. The purpose of the experiments was to evaluate the effectiveness of the Vehicle-to-Vehicle communication system and the decision-making algorithm in preventing collisions.

The system was tested under different vehicle interaction scenarios to observe how the vehicles communicated and coordinated their movement. The results demonstrate that the proposed system successfully enables vehicles to exchange information and make cooperative decisions in real time.[28,16]

7.1 Experimental Setup

The experimental setup consisted of miniature autonomous vehicles placed in a controlled environment to simulate different traffic situations. Each vehicle was powered by an ESP32 microcontroller that handled sensor input, Bluetooth communication, and motor control.

Table 7.1: System Performance Evaluation Results

Test Case	Scenario Description	System Response	Result
Test Case 1	Vehicles approaching from opposite directions	Both vehicles detect each other and stop	Collision avoided
Test Case 2	Vehicles moving in the same direction	Rear vehicle detects front vehicle and slows down	Safe distance maintained
Test Case 3	Vehicles approaching intersection from perpendicular directions	Vehicle with lower ID moves first	Coordinated movement achieved

7.2 System Performance

The system demonstrated stable operation during testing. The ESP32 microcontroller successfully processed sensor data and communication messages while controlling the movement of the vehicles through the motor driver module. Bluetooth communication between vehicles remained consistent within the operating range, enabling reliable data exchange.

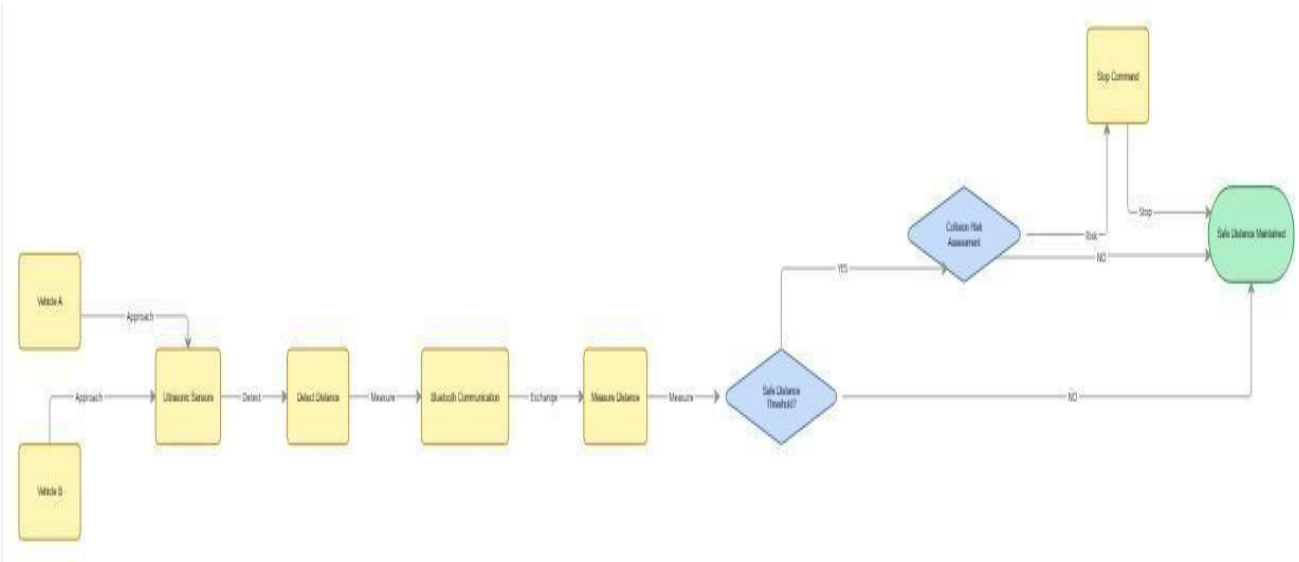

Figure 7.2: Collision Avoidance Demonstration

7.3 Collision Avoidance Performance

The collision avoidance capability of the system was evaluated using three test cases.

✓ **Test Case 1 – Opposite Direction Movement**

When two vehicles approached each other from opposite directions, the system detected the presence of the other vehicle and adjusted movement to avoid collision.

✓ **Test Case 2 – Same Direction Movement**

When one vehicle followed another vehicle moving in the same direction, the following vehicle detected the leading vehicle and reduced its speed to maintain a safe distance.

✓ **Test Case 3 – Intersection Scenario**

When vehicles approached an intersection from perpendicular directions, the system determined vehicle priority and allowed one vehicle to proceed while the other vehicle waited.

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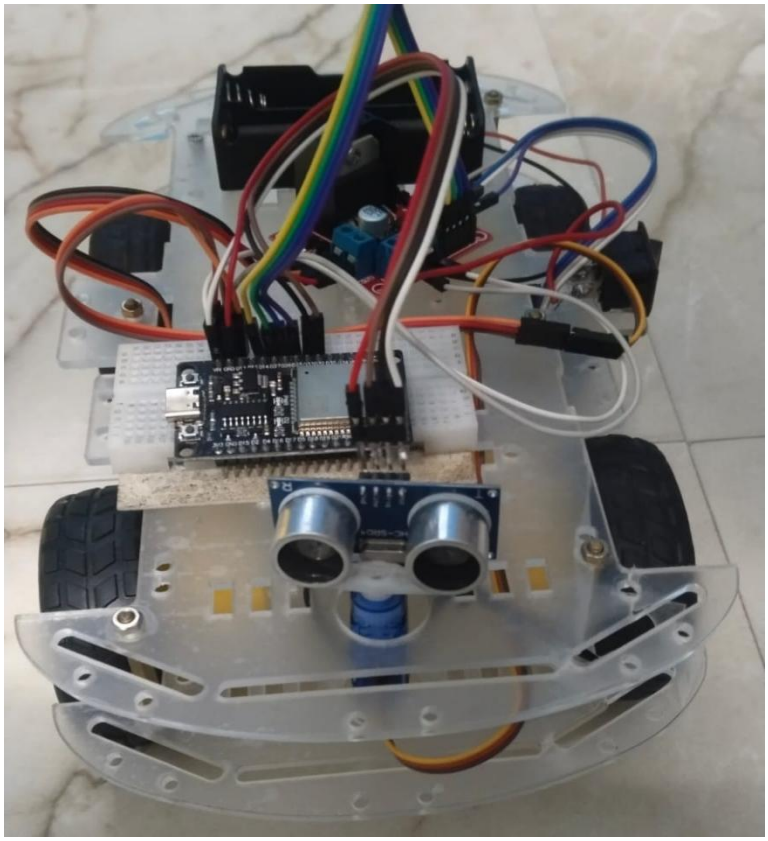
7.4 Work Efficiency

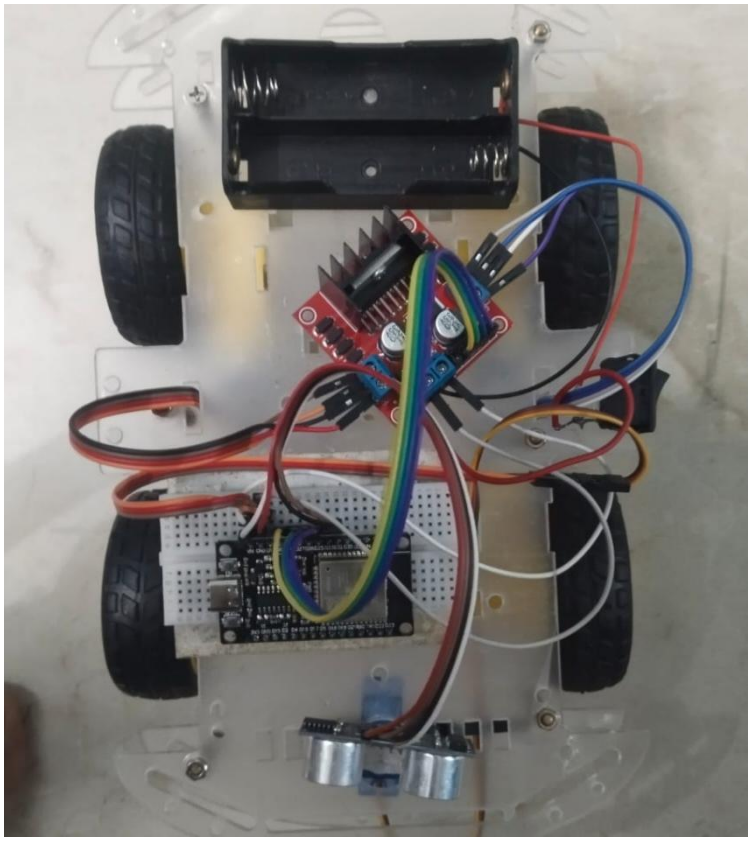
The developed system demonstrated efficient performance during testing. The ESP32 microcontroller successfully processed sensor data and Bluetooth communication simultaneously without delays. The vehicles were able to detect nearby vehicles, exchange information, and respond quickly according to the decision algorithm. The system maintained safe distances and prevented collisions in all three tested scenarios. Overall, the integration of sensing, communication, and control ensured smooth and reliable operation of the autonomous vehicle system.[12,25]

7.5 Result Summary

The experimental results confirm that the proposed system successfully integrates sensing, communication, and control to achieve cooperative vehicle behaviour. The ESP32 microcontroller efficiently handled sensor processing and Bluetooth communication, while the decision algorithm ensured safe vehicle movement.The system performed reliably in all tested scenarios, maintaining safe distances between vehicles and preventing collisions. These results demonstrate the feasibility of implementing autonomous vehicle coordination using low-cost embedded systems and wireless communication technology

7.6 Project Photos:





7.7 Simulation Results:

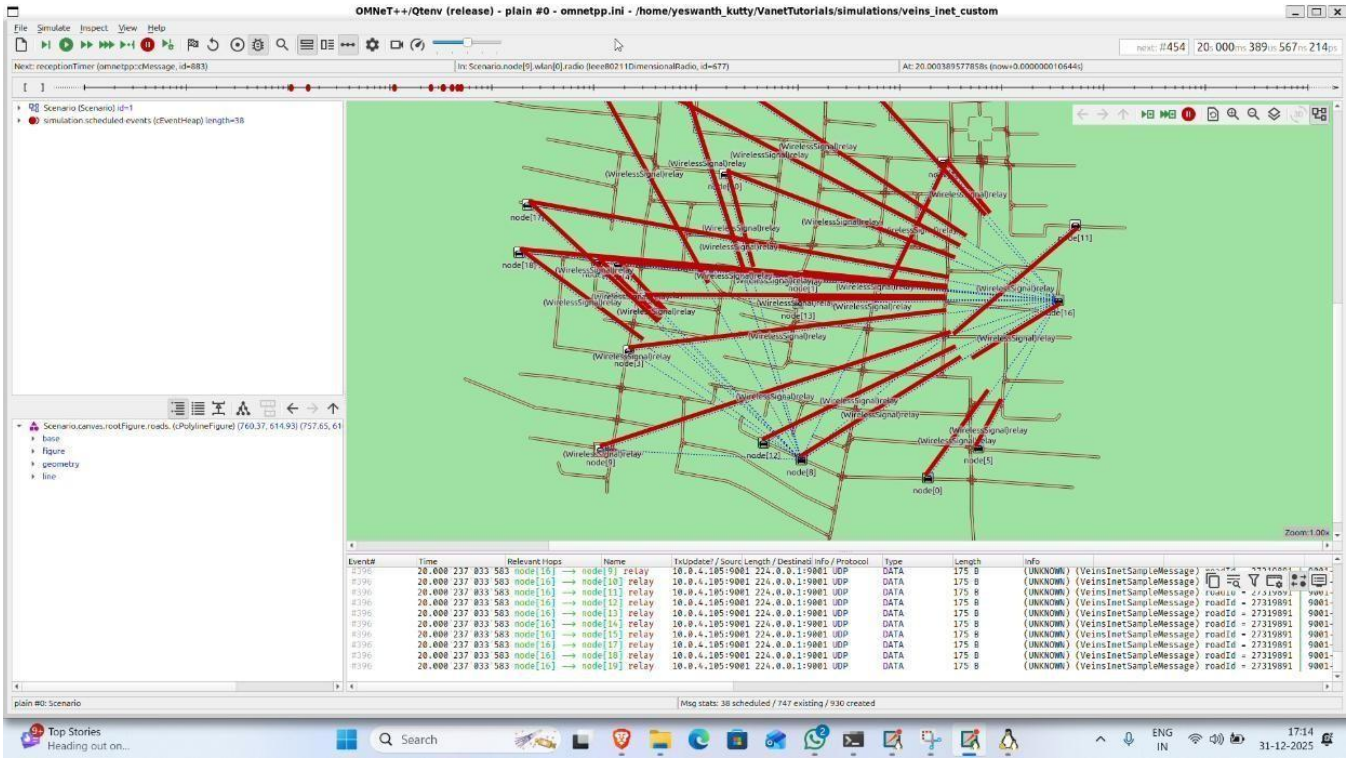


Fig 7.7.1: VEINS simulation environment showing V2V communication and intersection negotiation.

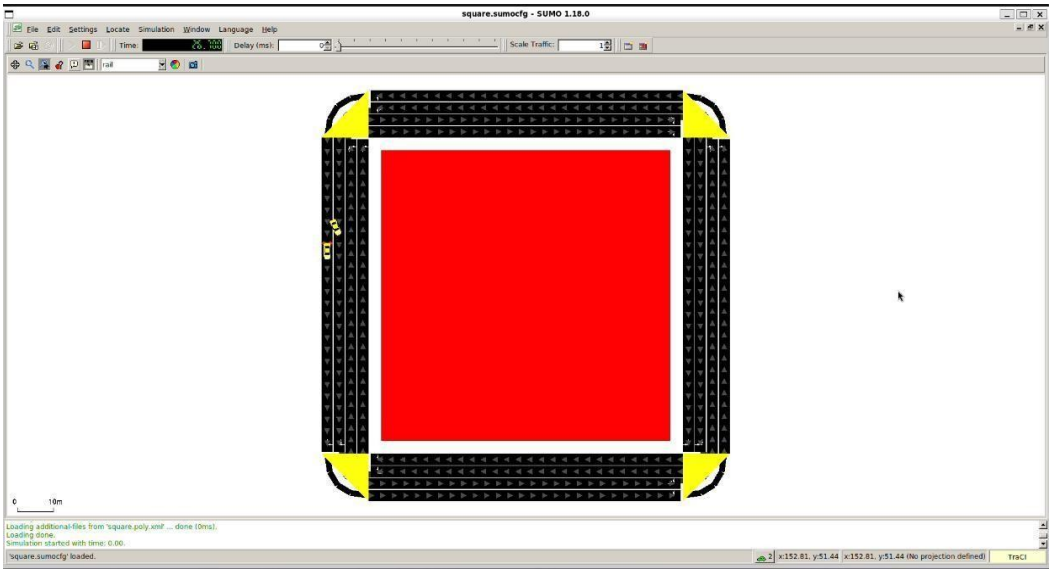


Fig 7.7.2 Collision Avoidance Warn drivers if another vehicle is approaching at high speed. Alert if a vehicle suddenly brakes

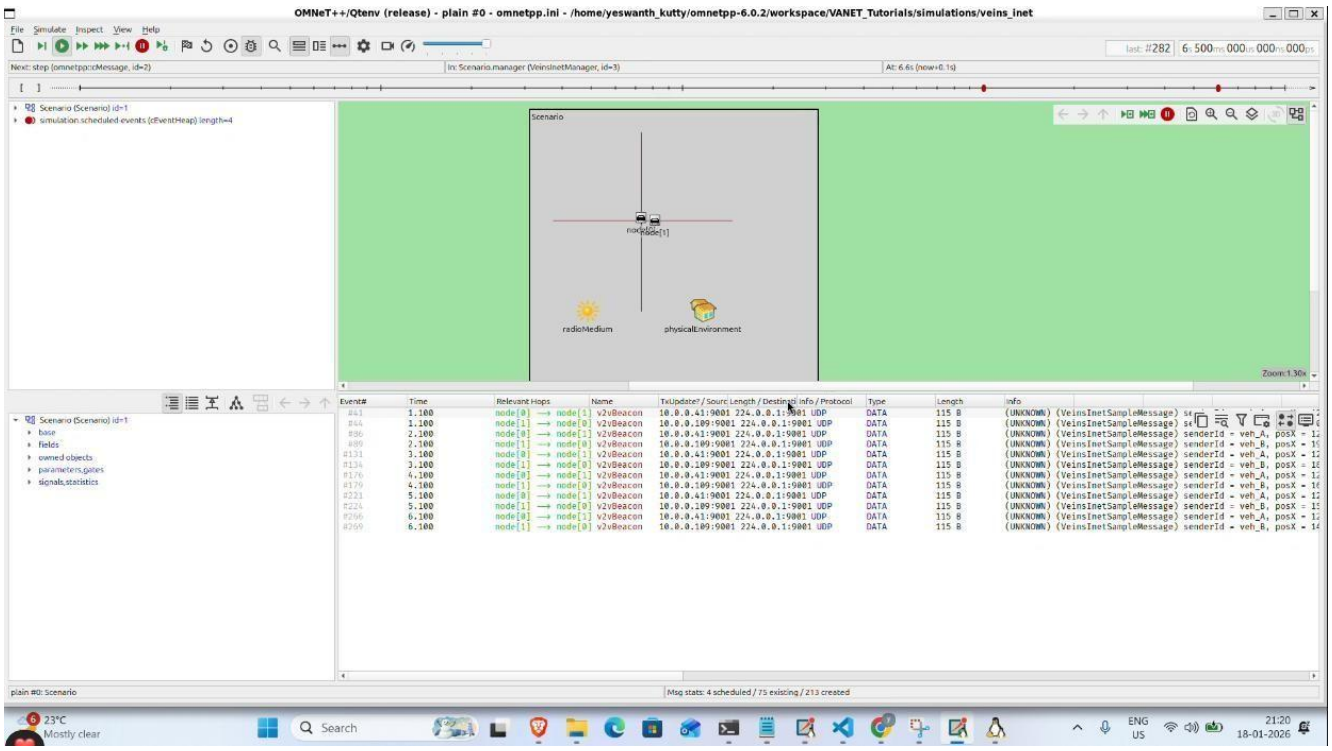


Fig 7.7.3 Intersection Safety Avoids Crashes at Junctions even when visibility is limited

CHAPTER 8
DISCUSSION

This chapter discusses the performance and significance of the developed autonomous vehicle system. The system was designed to demonstrate how vehicles can communicate with each other and make cooperative decisions to avoid collisions. The discussion focuses on interpreting the experimental results, evaluating the system's practical importance, and identifying possible limitations and improvements.

The results obtained from the testing scenarios show that the integration of sensors, wireless communication, and embedded control algorithms can successfully enable vehicles to coordinate their movement in different traffic situations[19,21].

8.1 Interpretation of Results

The experimental results indicate that the proposed system successfully implemented Vehicle-to-Vehicle communication and cooperative decision making. The ESP32 microcontroller effectively processed sensor inputs and communication data to determine vehicle actions. During testing, vehicles were able to detect nearby vehicles using ultrasonic sensors and exchange information using Bluetooth communication.

Based on the decision algorithm, vehicles adjusted their movement by slowing down, stopping, or continuing forward. The results demonstrate that the system can successfully perform basic autonomous decision-making tasks.

8.2 Significance

The proposed system demonstrates the practical application of embedded systems and wireless communication in intelligent transportation technologies. By enabling vehicles to communicate and cooperate, the system reduces the chances of collisions and improves overall traffic safety. The use of the ESP32 microcontroller with built-in Bluetooth communication provides a low-cost and efficient platform for implementing Vehicle-to-Vehicle communication. The system also demonstrates how simple rule-based algorithms can be used to achieve effective collision avoidance in small-scale autonomous vehicle models.

8.3 Comparison with Baseline

The developed system was compared with a baseline scenario where vehicles operate without communication or coordination. In the baseline case, vehicles rely only on manual control or basic obstacle detection, which can lead to delayed reactions and possible collisions. In contrast, the proposed system uses ESP32- based Bluetooth communication and sensor data to enable vehicles to exchange information and coordinate their movement.

8.4 Practical Relevance

1) The concepts demonstrated in this project have practical relevance in real- world transportation systems. Modern intelligent transportation systems rely on communication between vehicles to improve safety and traffic efficiency.
2) The project illustrates how vehicles can share information and make decisions collectively rather than operating independently. Such systems can be extended to
3) larger vehicle networks where communication and coordination play a critical role in preventing accidents.
4) Although this project is implemented using miniature vehicles, the underlying

principles are similar to those used in real autonomous vehicle technologies.[7,21]

8.5 Limitations

Despite the successful operation of the system, there are certain limitations that must be considered. The communication range of Bluetooth is limited compared to other wireless communication technologies used in real autonomous vehicle systems. The system also relies on simple rule-based decision algorithms, which may not handle highly complex traffic situations. Additionally, the testing environment was controlled and may not fully represent real-world traffic conditions.

These limitations indicate areas where further improvements can be made in future research.

8.6 Future Work Recommendations

Future work can focus on improving the system by integrating additional sensors and advanced decision-making algorithms. Technologies such as camera- based vision systems or LiDAR sensors could enhance environmental perception.

The communication system could also be upgraded to support longer communication ranges and faster data exchange. Implementing artificial intelligence or machine learning techniques may allow vehicles to make more advanced decisions in complex traffic scenarios.

These improvements could expand the capabilities of the system and make it more suitable for real-world autonomous transportation application.[19]

CHAPTER 9
CONCLUSION

9.1 Key Outcomes

The project successfully achieved the objectives of designing and implementing a miniature autonomous vehicle system capable of cooperative decision-making. The ESP32 microcontroller effectively managed sensor inputs, Bluetooth communication, and motor control. The system demonstrated reliable performance in three different scenarios: vehicles moving in opposite directions, vehicles moving in the same direction, and vehicles approaching an intersection from perpendicular directions. In each scenario, the vehicles were able to communicate and adjust their movement to avoid collisions.

9.2 Advantages

1) The proposed system offers several advantages. The use of the ESP32 microcontroller provides an integrated platform with built-in Bluetooth communication, reducing hardware complexity. The system uses simple sensors and rule-based algorithms, making it cost-effective and easy to implement.
2) The modular design allows the system to be expanded or modified for future research. The project also provides a practical demonstration of how communication between vehicles can improve safety and coordination.
3) Overall, the system demonstrates the feasibility of implementing cooperative autonomous vehicle behavior using compact embedded systems.

9.3 Future Scope

1) Although the system successfully demonstrates basic Vehicle-to-Vehicle communication and collision avoidance, there are several opportunities for further improvement.
2) Future developments could include integrating additional sensors such as cameras or LiDAR to improve environmental perception. The communication system could also be enhanced using more advanced wireless technologies with greater range and reliability.
3) More advanced decision-making algorithms, including artificial intelligence and machine learning techniques, could be implemented to handle more complex traffic scenarios. These improvements could extend the capabilities of the system and make it more suitable for real-world intelligent transportation applications.

9.4 Concluding Remarks

The project successfully demonstrates a miniature autonomous vehicle system capable of communication and cooperative decision-making. By combining sensing, Bluetooth communication, and embedded control, the system provides a practical model for studying intelligent transportation technologies.

The results highlight the importance of Vehicle-to-Vehicle communication in improving traffic safety and coordination. The project serves as a foundation for further research in autonomous systems, embedded communication networks, and smart mobility solutions.

APPENDIX

Appendix A: Hardware Components and Specifications

The mini autonomous vehicle system is built using compact and cost-effective embedded components. The primary hardware used in the system includes:

- ESP32 Microcontroller

Acts as the central processing unit. It handles sensor data processing, Bluetooth communication, and execution of the decision-making algorithm.

- Ultrasonic Sensor (HC-SR04)

Used for distance measurement and obstacle detection.

- Range: 2 cm – 400 cm
- Accuracy: ±3 mm
- Motor Driver Module (L298N)

Controls direction and speed of DC motors using PWM signals.

- DC Motors and Chassis

Provide motion and structural support for the vehicle.

- Power Supply Unit

Supplies required voltage for the entire system.

Appendix B: Algorithm Flow (Simplified)

The system follows a continuous loop-based embedded control logic:

1. Initialize ESP32, sensors, and Bluetooth module
2. Measure distance using ultrasonic sensor
3. Transmit vehicle data (ID, distance, status)
4. Receive data from nearby vehicles
5. Process data using decision algorithm
6. Apply rules:

- If obstacle detected → Stop / Slow down
- If path is clear → Move forward
- At intersection → Apply priority rule

7. Send control signals to motor driver
8. Repeat continuously

Appendix C: Communication Data Format

Field	**Description**	**Example**
Vehicle ID	Unique identifier	1
Direction	Movement direction	Forward
Distance	Measured distance	20 cm
Status	Vehicle state	Moving

Appendix D: Test Scenarios Summary

The system was evaluated using three real-time scenarios:

1. Opposite Direction Movement

Vehicles detect each other and stop to avoid collision.

2. Same Direction Movement

Rear vehicle maintains safe distance by adjusting speed.

3. Intersection Scenario

Vehicles communicate and decide priority for safe crossing.

Appendix E: Simulation Tools

The project uses simulation tools to validate performance:

- OMNeT++ – Network simulation
- SUMO (Simulation of Urban Mobility) – Traffic simulation

- VEINS Framework – Integration of communication and mobility

These tools help analyze vehicle behavior, communication efficiency, and collision avoidance performance.

Appendix F: Key Advantages of System

- Low-cost implementation using ESP32
- Real-time vehicle communication
- Simple and effective decision-making algorithm
- Scalable for future improvements
- Demonstrates core concepts of intelligent transportation systems

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PO & PSO ATTAINMENT

PO.No	Graduate Attribute	Attained	Justification
PO 1	**Engineering knowledge**	Yes	Applied embedded systems, electronics, and communication concepts
PO 2	**Problem analysis**	Yes	Intersection collision risk was clearly analyzed and addressed.
PO 3	**Design/Development of solutions**	Yes	Designed a complete autonomous V2V system solution.
PO 4	**Conduct investigations of complex problems**	Yes	Used simulation and real testing for system evaluation.
PO 5	**Modern Tool Usage**	Yes	Used Arduino, VEINS, OMNeT++, SUMO, and wireless tools.
PO 6	**The Engineer and society**	Yes	Contributes to public safety and accident prevention.
PO 7	**Environment and Sustainability**	Yes	Used low-power and energy efficient components.

PO 8	**Ethics**	Yes	Focused on safety, reliability, and responsible automation..
PO 9	**Individual and teamwork**	Yes	Achieved through teamwork and collaboration.
PO 10	**Communication**	Yes	Effectively documented architecture design, performance results, and comparative analysis through technical reports, presentation slides, and graphical results.
PO 11	**Project management and finance**	Yes	Completed within cost and time constraints
PO 12	**Life-long learning**	Yes	Encouraged continuous learning in autonomous systems.

PSO.No	Graduate Attribute	Attained	Justification
PSO 1	To analyze, design, and develop electronic and intelligent systems using core concepts of Electronics and Communication Engineering.	YES	The project applies electronics, embedded systems, control systems, and communication principles to build an autonomous vehicle system.
PSO 2	To create innovative solutions for real-world Considerations by adapting modern tools, automation technologies, and interdisciplinary approaches.	YES	The system integrates embedded systems, wireless communication, simulation modeling, and intelligent decision-making for sustainable smart mobility solutions..